

MVP Architecture RAG and Technical Feasibility

IP SAKTI Sahayak | SIH 2026 | Research Pack for Anshuman

Simple Hinglish notes for team explanation and implementation planning

1 Simple Explanation

Hamare MVP ko ek source backed helper samjho. User apna product ya question batayega. Website request FastAPI backend ko bhejegi. Backend pehle product type aur India or International jurisdiction check karega. Jab verified documents ka corpus ready hoga, system unhi documents ke relevant parts dhoondhega aur answer ke saath citation, confidence aur disclaimer dega.

Aaj ki sachchai: React Vite UI, FastAPI health route, chat API contract, language aur jurisdiction controls available hain. RAG retrieval, verified corpus, citations, confidence aur refusal abhi build karne hain. Isliye presentation mein inhe “MVP next” bolna hai, “already working” nahi.

2 Architecture Diagram

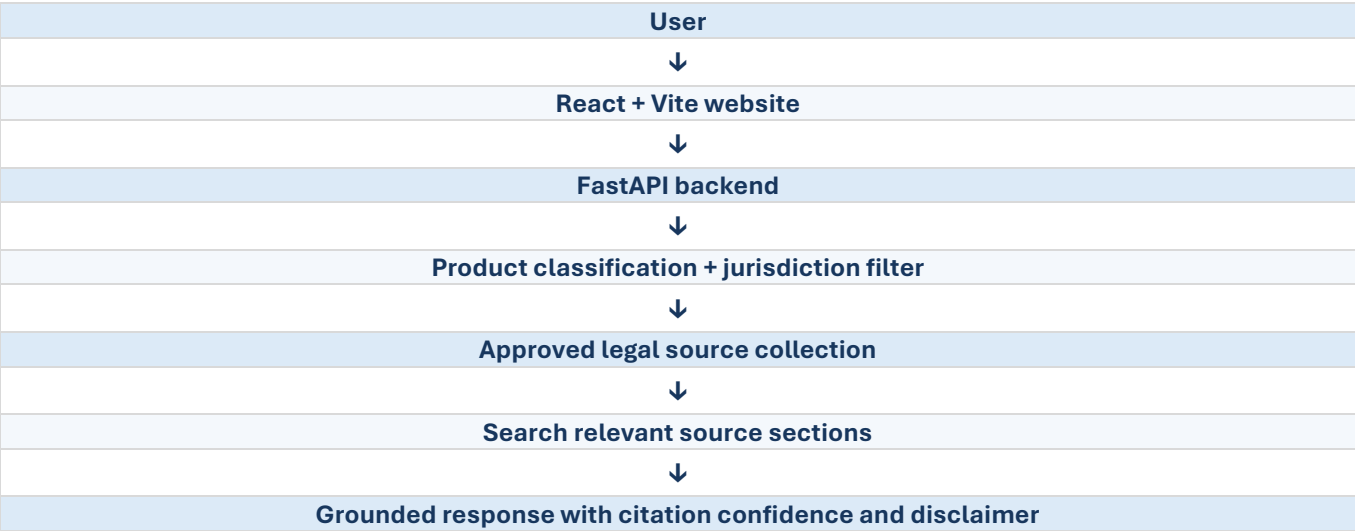


Figure 1. Target MVP architecture. The lower retrieval stages are planned, not yet implemented.

3 Components in Easy Language

Part	Simple kaam	MVP truth
React + Vite	User ka chat page, language and jurisdiction choice	Implemented UI layer
FastAPI + Python	Request validate karna aur JSON response dena	Health route and chat contract implemented
Pydantic validation	Question blank na ho; language and jurisdiction allowed values mein ho	Current API contract ka part
Product classifier	Question ko medicine, food, cosmetic, formulation etc. ke path par bhejna	Planned refinement
Approved corpus	Sirf checked official laws, rules and treaty documents	MVP next; source register verification needed
ChromaDB vector store	Source sections ko searchable form mein rakhna	Proposed local store
Sentence Transformers	Question aur source sections ka meaning compare karne ke embeddings	Proposed embedding layer

Answer guardrail	Citation na mile to safe refusal; legal advice nahi	MVP next
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4 RAG Workflow Diagram

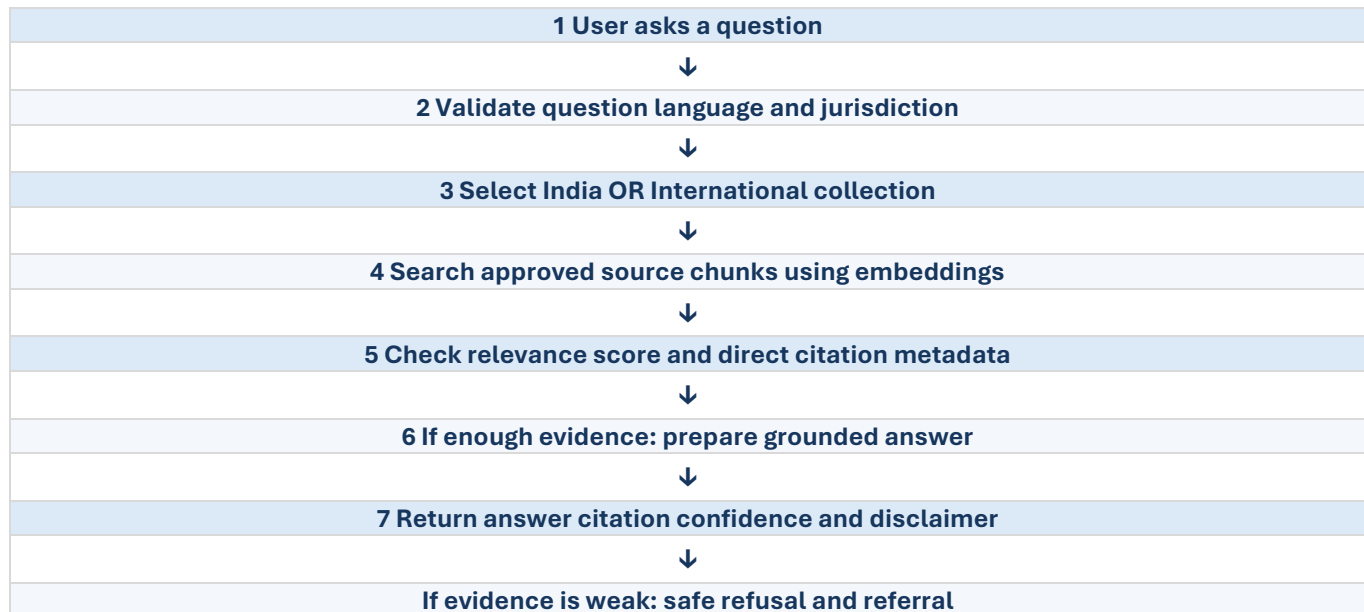


Figure 2. RAG means search the approved books first, then answer only from what was found.

5 RAG Concepts KISS Version

Term	Bilkul simple meaning	Why it matters
Knowledge corpus	Verified official documents ka controlled folder	Random web page ya model memory se answer nahi
Chunking	Long PDF ko chhote readable pieces mein todna	Exact section easily milta hai; context manageable rehta hai
Embedding	Text meaning ka numeric fingerprint	Same meaning wale question and source ko match karta hai
Vector search	Fingerprint se sabse relevant chunks dhoondhna	Keyword same na ho tab bhi useful match mil sakta hai
Citation	Exact Act rule treaty title plus section page or stable heading	User team aur faculty source check kar sakte hain
Confidence	Evidence kitna direct aur strong hai uska simple label	High / Moderate / Low; legal certainty ka claim nahi
Safe refusal	Direct authoritative evidence nahi mila to answer invent na karna	Hallucinated legal citation rokta hai

6 India and International Sources Separate Kyun

Same product ka rule India aur kisi other country mein alag ho sakta hai. India collection mein Indian Acts, Rules, Regulations aur official Indian authorities honge. International collection mein treaties aur WIPO/WTO/CBD material hoga. Ek retrieval step mein dono collections mix nahi honge. User explicitly International choose kare to treaty awareness mil sakti hai, lekin destination country registration, labelling ya market entry ke liye relevant regulator ya qualified professional ko consult karna hoga.

India collection	International collection	Non negotiable rule
Indian Acts, Rules, FSSAI / AYUSH / IP India / NBA official sources	Treaties and WIPO WTO CBD official sources	No combined retrieval answer

India jurisdiction metadata and source section	International jurisdiction metadata and treaty article	Every citation shows its jurisdiction
India first product guidance	Treaty awareness and export routing only	No country by country legal advice

7 Status Table

Status Items	
Implemented	React/Vite UI, FastAPI health route, chat API contract, language and jurisdiction controls
MVP next	Verified source corpus, text extraction, retrieval, citations, confidence, and safe refusal
Future	Voice, Bhashini integration, knowledge graph, advanced export guidance, and user accounts

8 Five MVP Test Questions

#	Test question	Expected safe MVP behaviour
1	Can I patent a classical Ayurvedic formulation in India?	India collection se direct source section retrieve ho only then cited informational answer; otherwise refusal.
2	I want to sell an Ashwagandha drink. Is it food or medicine?	Product classification questions; explain it needs classification and point to verified India source; no final approval claim.
3	Can I use Neem from India for a commercial product?	Biodiversity / ABS source route if verified; show citation; recommend authority or professional where facts are missing.
4	Can I register my herbal brand name in India?	Trademark awareness and official route; do not promise registration or clearance.
5	Can I sell my Ayurvedic product in the EU?	International mode: name relevant treaty/authority only if sourced; refuse destination market clearance and refer to EU regulator / expert.

9 What I Need To Learn

Priority	Topic	One line learning goal
Day 1	React request flow	Understand question, language and jurisdiction frontend se backend kaise jaate hain.
Day 1	FastAPI + Pydantic	POST request, response model, validation error aur /docs try karo.
Day 2	PDF text extraction	PyMuPDF se PDF text nikalo; source title, page and section metadata preserve karo.
Day 2	Chunking	Start with small meaningful chunks and overlap; headings/pages ko metadata mein rakho.
Day 3	Embeddings + ChromaDB	One query ko vector bana kar top matching chunks dekho.
Day 4	Grounded response	Retrieved context ke bina answer na dene wali prompt and safe refusal banao.
Day 5	Citations + confidence	Citation metadata render karo; confidence score ko simple High Moderate Low labels mein map karo.
Day 6	Test and demo	Five test questions run karo; wrong or weak retrieval ko refusal mein convert karo.

10 Install Plan for ML AI Part

Pehle Python 3.11+ virtual environment use karo. Full heavy model setup se shuru mat karo. Small verified corpus par local proof of concept banao. Team ke backend folder ke README / requirements file ko final source of truth rakho.

Need	Why	Start with
Python 3.11+ and venv	Isolated project dependencies	python -m venv .venv
FastAPI + Uvicorn	Service and local API testing	pip install fastapi uvicorn
PyMuPDF	Official PDFs se text extraction	pip install pymupdf
Sentence Transformers	Question and chunk embeddings	pip install sentence-transformers
ChromaDB	Local vector search store	pip install chromadb
pytest optional	API and retrieval test cases	pip install pytest

Important: Install karne se pehle backend teammate ke saath versions freeze karo. Corpus documents ko verified status ke bina ChromaDB mein load mat karo. API key based LLM connector later add hoga; first demo retrieval plus cited safe response se prove karo.

11 Three Minute Speaking Notes

- Hamare MVP ka goal legal advice dena nahi hai; official source backed information ko simple language mein organise karna hai.
- Abhi UI aur FastAPI API contract ready hai. RAG pipeline ka corpus, search, citations aur refusal next implementation phase hai.
- RAG ko open book exam samjho: pehle approved document ke relevant pages search honge, fir unhi par based answer banega.
- India aur International sources alag rahenge because rules aur product treatment jurisdiction ke hisaab se change hote hain.
- Agar relevant verified citation nahi milti, system answer guess nahi karega. It will say it cannot confirm from an authoritative source and ask the user to consult the right authority or professional.

12 Risks and Practical Mitigation

Risk	Why it matters	Practical mitigation
Unverified document	Wrong or outdated legal text could be retrieved	Use Source Register; ingest only verified items
Mixed jurisdiction	India law and treaty context may sound like one rule	Separate collections and a mandatory jurisdiction filter
Weak match	Vector search can return related but not direct text	Threshold, citation validation and safe refusal
Overclaim in demo	Faculty may assume future work is live	Use Implemented / MVP next / Future status labels
Time constraint	Six days is limited for a broad corpus	Start with 3 to 5 verified India sources and five tests

13 Official Starting Sources

Source	Use in this work	URL
Project Architecture PRD Rules and Phases	Current implementation status, scope and guardrails	Local project docs
FastAPI documentation	Python API routes, validation and interactive API docs	https://fastapi.tiangolo.com/
ChromaDB documentation	Vector database concepts and collections	https://docs.trychroma.com/
Sentence Transformers documentation	Embedding model and semantic similarity concepts	https://sbert.net/

Accessed: 04 September 2026. For legal corpus documents, use only official issuing-authority links and record verification status, page/section and jurisdiction in SOURCE_REGISTER.md before ingestion.